

Machine Learning-Based Phishing Website Detection

Comparative Analysis of Logistic Regression, Random Forest, and Neural Networks

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1. Executive Summary

Phishing websites continue to be one of the most prevalent cyber threats, enabling attackers to steal credentials, distribute malware, and compromise sensitive information through deceptive web content. Traditional blacklist-based detection methods often struggle to identify newly created phishing domains, making machine learning a practical approach for detecting malicious websites based on their behavioral and structural characteristics.

This project evaluates three supervised machine learning algorithms for phishing website detection using the same feature-engineered dataset. The models include Logistic Regression, Random Forest, and a Multi-Layer Perceptron (Neural Network). Each model was trained using identical preprocessing steps to ensure a fair comparison. Their performance was evaluated using accuracy, ROC-AUC, confusion matrices, cross-validation, and classification metrics.

Among the evaluated algorithms, the Random Forest classifier achieved the strongest overall performance, demonstrating the highest predictive accuracy and ROC-AUC while maintaining robust generalization. The project illustrates how machine learning can automate phishing detection and support enterprise security operations by improving the identification of malicious websites.

2. Project Objectives

The primary objectives of this project were to:

- Develop supervised machine learning models capable of distinguishing phishing websites from legitimate websites.
- Compare the performance of multiple classification algorithms under identical experimental conditions.
- Evaluate each model using industry-standard machine learning metrics.
- Identify the most suitable algorithm for practical phishing website detection.

- Demonstrate how machine learning techniques can support cybersecurity threat detection and automated decision-making.
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3. Dataset and Feature Engineering

The project utilizes a structured phishing website dataset containing thousands of labeled website samples represented by engineered numerical features. Each record corresponds to a single website and is classified as either legitimate or phishing.

Rather than analyzing raw HTML pages directly, the dataset contains extracted features describing website behavior and structure. Examples include:

- Number of external JavaScript files
- Number of external resources
- Presence of inline scripts
- Event handler usage
- Library count
- HTML structural characteristics
- Script-related attributes
- Additional engineered indicators associated with phishing behavior

The target variable represents the website class, where each observation is labeled as either legitimate or phishing.

Using engineered features significantly reduces computational complexity while allowing machine learning algorithms to identify patterns commonly associated with malicious websites.

4. Data Preprocessing

Before model training, the dataset underwent several preprocessing steps to ensure data quality and consistency.

The dataset was imported using the Pandas library and separated into independent variables (X) and the target variable (y). A train-test split was performed to divide the

dataset into training and testing subsets while preserving the distribution of phishing and legitimate samples.

For algorithms sensitive to feature magnitude, particularly Logistic Regression and the Neural Network, feature scaling was performed using StandardScaler. Standardization transforms each feature to have a mean of zero and a standard deviation of one, improving convergence during optimization and preventing features with larger numeric ranges from dominating the learning process.

Cross-validation techniques were also applied during model evaluation to reduce overfitting and provide a more reliable estimate of generalization performance.

5. Machine Learning Models

5.1 Logistic Regression

Logistic Regression served as the baseline classification model for the study. Although mathematically simple, it remains widely used in cybersecurity due to its interpretability and computational efficiency.

The model estimates the probability that a website belongs to the phishing class using the logistic (sigmoid) function. During training, model parameters are optimized by minimizing the logistic loss function through iterative optimization.

The implementation included:

- Feature standardization
- Model fitting using Scikit-learn
- Classification prediction
- Confusion matrix generation
- ROC curve analysis
- Cross-validation

The Logistic Regression model produced satisfactory performance and established a baseline for comparison with more advanced algorithms. However, because it assumes a linear decision boundary, it is less effective at modeling complex relationships within phishing datasets.

Approximate Performance:

- Accuracy: ~82%
 - ROC-AUC: ~0.85
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5.2 Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions through majority voting. Each tree is trained using a random subset of the training data and a random subset of features, significantly reducing variance and improving robustness.

Several configurations were evaluated by varying the number of decision trees to determine the optimal balance between performance and computational complexity.

Model tuning included experiments with:

- 100 trees
- 200 trees
- 300 trees
- 500 trees
- 700 trees
- 900 trees

Feature importance scores generated by the Random Forest model provided valuable insight into which website characteristics contributed most significantly to phishing detection.

The implementation also included:

- Hyperparameter tuning
- Cross-validation
- ROC analysis
- Confusion matrix evaluation
- Feature importance visualization

Among all evaluated algorithms, Random Forest achieved the highest predictive performance.

Approximate Performance:

- Accuracy: ~92%
- ROC-AUC: ~0.95

The ensemble nature of Random Forest allows it to capture complex nonlinear relationships that simpler models cannot effectively represent.

5.3 Multi-Layer Perceptron (Neural Network)

A Multi-Layer Perceptron (MLP) classifier was implemented to evaluate the effectiveness of neural networks for phishing website classification.

The neural network consists of interconnected neurons organized into input, hidden, and output layers. During training, the model adjusts network weights using backpropagation to minimize prediction error.

The implementation included:

- Feature standardization
- Hidden layer optimization
- Iterative gradient-based learning
- Performance evaluation
- ROC analysis
- Confusion matrix generation

Although the neural network successfully learned nonlinear decision boundaries, its performance remained below that of the optimized Random Forest classifier.

Approximate Performance:

- Accuracy: ~86%
- ROC-AUC: ~0.86

This outcome suggests that, for the selected dataset and feature set, the ensemble learning approach generalized more effectively than the neural network architecture.

6. Model Performance Comparison

All three machine learning algorithms were evaluated using identical datasets and preprocessing procedures.

Model	Approximate Accuracy	ROC-AUC	Characteristics
Logistic Regression	~82%	~0.85	Simple, interpretable baseline model
Random Forest	~92%	~0.95	Highest overall performance and robustness
Neural Network	~86%	~0.86	Captures nonlinear relationships but requires greater computational resources

The Random Forest classifier consistently outperformed the remaining algorithms across multiple evaluation metrics. Its ensemble architecture effectively reduced overfitting while maintaining high predictive capability, making it the most suitable model for phishing website detection within this project.

7. Security Implications

Machine learning has become an increasingly valuable component of modern cybersecurity defenses. The techniques demonstrated in this project have practical applications across multiple security domains, including:

- Email security gateways
- Secure web gateways
- Browser-based phishing protection
- Enterprise web filtering
- Threat intelligence platforms
- Security Operations Centers (SOC)
- Automated phishing detection pipelines

By learning patterns associated with phishing websites rather than relying solely on static blacklists, machine learning models can identify previously unseen malicious websites and improve detection capabilities against evolving threats.

The ability to automatically classify suspicious websites also reduces analyst workload and enables faster incident response.

8. Conclusion

This project demonstrated the practical application of supervised machine learning techniques for phishing website detection through a comparative evaluation of Logistic Regression, Random Forest, and Multi-Layer Perceptron classifiers.

Using identical preprocessing procedures and evaluation metrics ensured an objective comparison between models. While Logistic Regression provided a strong interpretable baseline and the Neural Network successfully captured nonlinear feature relationships, the Random Forest classifier consistently achieved the highest predictive performance. Its superior accuracy, ROC-AUC, and robustness indicate that ensemble learning is particularly effective for phishing website classification.

Overall, the project highlights how machine learning can strengthen cybersecurity defenses by automating website classification, improving phishing detection accuracy, and supporting enterprise security operations with scalable, data-driven decision-making.